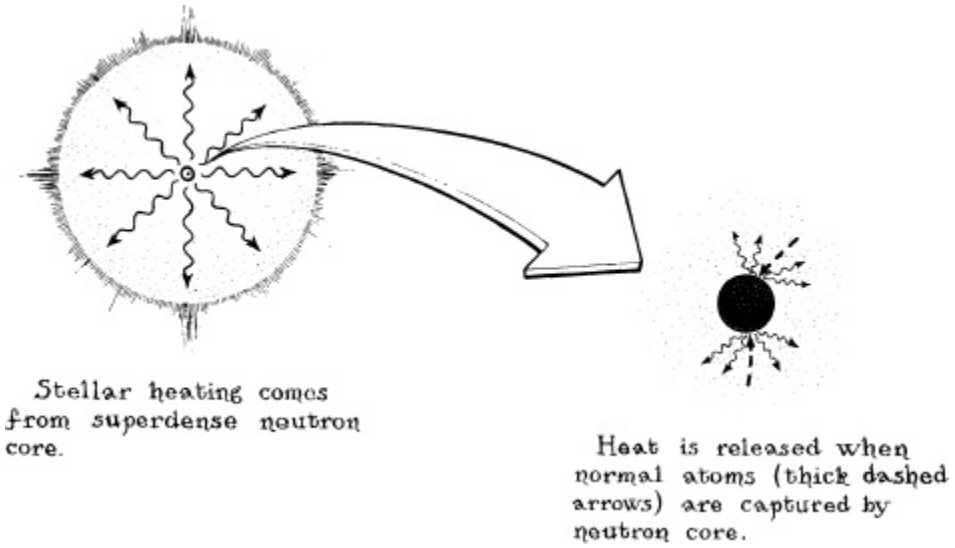


5.4 Lev Landau's speculation as to the origin of the energy that keeps a normal star hot.

Box 5.3

Nuclear Burning (Fusion) Contrasted with I Ordinary Burning

Ordinary burning is a *chemical reaction.* In chemical reactions, atoms get combined into molecules, where they share their electron clouds with each other; the electron clouds hold the molecules together. Nuclear burning is a *nuclear reaction.* In nuclear burning, atomic nuclei get fused together (*nuclear fusion*) to form more massive atomic nuclei; the nuclear force holds the more massive nuclei together.

The following diagram shows an example of ordinary burning: the burning of hydrogen to produce water (an explosively powerful form of burning that is used to power some rockets that lift payloads into space). Two hydrogen atoms combine with an oxygen atom to form a water molecule. In the water molecule, the hydrogen and oxygen atoms share their electron clouds with each other, but do not share their atomic nuclei.